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Exploring Speech Emotion Recognition with MLP Classifier: A Comprehensive Study on the CREMA-D Dataset

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Abstract: Speech emotion recognition is a pivotal aspect of human-computer interaction and affective computing. This research delves into the application of the Multi-Layer Perceptron (MLP) Classifier for emotion recognition within speech, focusing on the widely used CREMA-D dataset. Our investigation covers meticulous pre-processing methods, feature extraction techniques, and model configuration. Raw audio data undergoes preprocessing, with acoustic features like Mel-frequency cepstral coefficients (MFCCs) and pitch-related parameters extracted. The MLP Classifier, a feedforward neural network, is employed to categorize emotions in speech samples. Our results reveal competitive performance in recognizing a spectrum of emotions, including happiness, sadness, anger, and neutrality within the CREMA-D dataset. Our study examines the influence of di-verse hyperparameters, training strategies, and model architectures on emotion recognition accuracy. Cross-validation ensures the reliability of our findings, emphasizing the relevance of dataset selection and the potential for advancing speech emotion recognition research. This research aids in enhancing human-computer interaction systems, sentiment analysis, and emotional well-being monitoring applications, while also contributing to the discourse on optimizing this technology.

Keywords— Speech Emotion Recognition, MLP Classifier, CREMA-D Dataset, Emotion Classification, Acoustic Feature Extraction.

1. INTRODUCTION

Speech, as a medium for conveying human emotions, carries a wealth of valuable information that can be harnessed for diverse applications in human-computer interaction, sentiment analysis, and affective computing. Recognizing and understanding emotions from speech signals is a burgeoning field with implications ranging from improving user experiences in human-computer interaction to enhancing mental health monitoring systems. In this context, the present research delves into the realm of speech emotion recognition, a technology that offers remarkable potential.

This paper explores the application of the Multi-Layer Perceptron (MLP) Classifier in recognizing emotions within speech samples. We focus our attention on the use of the renowned CREMA-D dataset, a resource encompassing a diverse range of emotional expressions in speech [1]. The primary objective of this study is to provide a comprehensive analysis, from pre-processing raw audio data to feature extraction and MLP Classifier model configuration. The nuanced landscape [16] of speech emotion recognition, we aim to shed light on the intricate relationship between acoustic features and emotional states. We seek to uncover the efficacy of the MLP Classifier as a tool for this purpose and to understand how various factors, including hyperparameter settings, training strategies, and model architectures, impact its performance. The significance of dataset selection in this context is also underlined, as it plays a pivotal role in the generalization and applicability of the model [2].

Our research not only contributes to the advancement of speech emotion recognition technology but also underscores its real-world applications, from improving human-computer interaction systems to aiding in the monitoring of emotional well-being. It is within this exciting and ever-evolving realm that we embark on our comprehensive exploration, seeking to bridge the gap between spoken words and the profound emotions they convey.

The human voice, with its intricate auditory cues, stands as an invaluable reservoir of emotional cues. The domain of Speech Emotion Recognition (SER) has garnered remarkable attention, driven by its multifaceted practical implications. This study dives deep into SER, with a specific focus on the application of the Multi-Layer

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Perceptron (MLP) Classifier, a widely adopted neural network architecture, for decoding and categorizing emotional states embedded within spoken language.

SER exhibits substantial potential across a spectrum of applications, spanning human-computer interaction, customer sentiment analysis, and mental health monitoring. Its essence lies in its capacity to unveil the emotional subtleties within verbal expressions, enabling machines to comprehend and respond to human sentiments. The objective of this research is to investigate the efficacy of the MLP Classifier within the SER domain. Our motivation stems from the imperative need to leverage artificial intelligence in decoding emotions. Our research centers on the intricate details enshrined within the CREMA-D dataset, involving an exhaustive analysis of pre-processing techniques, feature extraction strategies, and model configurations vital for precise and resilient emotion recognition [3]. Our exploration of this dynamic field not only aims to enhance the MLP Classifier's performance but also contributes to the broader discourse surrounding SER. Our discoveries may potentially pave the way for emotionally intelligent technologies, ushering in new dimensions of interaction and understanding between humans and machines. This journey of exploration is firmly rooted in the belief that the human voice, with its emotional tapestry, is the gateway to unlocking more empathetic and responsive technological advancements.

II. Literature Survey

In this study, the research explores the potential of Neural Network (NN) models for recognizing speech emotions. Various NN models, including Radial Basis Function Network (RBFN), Multilayer Perceptron (MLP), and Probabilistic Neural Network (PNN), are extensively simulated to assess their Speech Emotion Recognition (SER) accuracy for emotions like anger, happiness, sadness, and boredom. The research uses efficient cepstral domain vocal tract system features, such as LPCC, MFCC, and PLP, extracted at a frame level and clustered into Vector Quantized (VQ) coefficients. The results reveal that PNN, utilizing LPCCVQ feature sets, achieves an impressive 83% accuracy, outperforming RBFN and MLP. LPCCVQ is identified as the most reliable feature set for characterizing speech emotions [4]

This article discusses the outcomes of the fifth Affective Behavior Analysis in-the-wild (ABAW) competition. The study investigates the use of pre-trained convolution-al networks from the EmotiEffNet family for frame-level feature extraction. The proposed approach combines a multi-layered perceptron and a Light AutoML-based classifier, with post-processing involving sequential frame smoothing. Results from experiments on the large-scale Aff-Wild2 database reveal significant improvements in macro-averaged F1-score for facial expression recognition, action unit detection, and concordance correlation coefficients for valence/arousal estimation compared to baseline methods [5].

In the realm of identifying emotional markers in human speech, this study addresses the challenge. While prior research primarily focused on predicting emotional states from speech, this investigation centers on analyzing affective vocal bursts. Leveraging self-supervised learning techniques, acoustic embeddings were extracted using wav2vec 2.0 variants for tasks related to High, Two, Culture, and Type. Notably, the evaluation of these acoustic embeddings revealed their potential superiority over traditional acoustic features [14] for affective vocal burst tasks. By assessing both types of features and applying them to a validation set with twenty different seeds, the study reported maximum and average scores along with their standard deviations. The outcomes of these evaluations significantly enhanced predictions for the test set when compared to previous studies [6].

Recent advancements in Speech Emotion Recognition (SER) have incorporated both acted and real-world datasets, raising questions about the uniformity and reliability of their annotations. The CMU-MOSEI video dataset, a prominent in-the-wild resource, employs "soft labels" on a Likert scale, often simplified to binary annotations. This study re-evaluated a small portion of two acted and two in-the-wild datasets, aiming to assess annotation strength, compare accuracy in both settings and establish an optimal threshold for CMU-MOSEI label binarization. The results revealed a substantial performance boost (7% increase in weighted average F1) by identifying a more suitable threshold, indicating that model architecture performs consistently across acted and in-the-wild data [7].

A. Introduction to Speech Emotion Recognition [8]:

Previous Research: Pioneering work by Ekman (1992) emphasized the existence of basic emotions. Recent Trends: Contemporary SER research has shifted towards machine learning and deep learning approaches.

B. Machine Learning and Deep Learning in SER[9]:

Previous Research: Papers like Deng et al. (2013) demonstrated the transition from traditional methods to machine learning for SER Recent Trends: More recent work by Zhang et al. (2020) has explored advanced deep learning techniques in SER, such as convolutional recurrent neural networks (CRNN).

C. Emotion Datasets:

Previous Research: Stropharia and Mihalcea (2007) introduced the idea of identifying emotions in text, paving the way for emotion datasets. Recent Trends: The CREMA-D dataset (Cao et al., 2014) has gained prominence in recent years due to its comprehensive collection of emotional speech data.

D. Deep Learning and SER

Previous Research: Kim and Lee (2016) illustrated the use of convolutional neural networks (CNNs) for emotion recognition in speech.

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Recent Trends: Deep learning models have continued to evolve with enhanced architectures, allowing better feature extraction and emotion classification.

D. CREMA-D Dataset:

Previous Research: Early studies did not have access to datasets like CREMA-D, which provides a diverse range of emotions in speech.

Recent Trends: Researchers now have the advantage of using the CREMA-D dataset for in-depth studies in SER.

E. Multilayer Perceptron's (MLPs) in Speech Emotion Recognition: Previous Research: Xu et al. (2016) explored the use of MLPs for voice activity detection.

Recent Trends: Your study investigates the application of MLPClassifiers in speech emotion recognition, contributing to the research landscape.

F. Feature Extraction for SER:

Previous Research: Eyben et al. (2010) introduced opens mile as a versatile audio feature extraction tool.

Recent Trends: Feature extraction tools like open smile continue to be valuable in modern SER, aiding in the extraction of relevant audio features.

G. SER Evaluation Metrics:

Previous Research: Schuller et al. (2013) introduced benchmark metrics for evaluating SER systems in the INTERSPEECH 2013 computational paralinguistics challenge.

Recent Trends: These benchmark metrics continue to be essential in assessing the performance of SER models.

H. State-of-the-Art SER Models:

Previous Research: Zhang et al. (2020) proposed models that utilize local and global patterns for speech emotion recognition, indicating progress in the field.

Recent Trends: Continual advancement in SER models has led to better accuracy and reliability.

I. Challenges and Future Directions:

Previous Research: Cowie et al. (2001) discussed the challenges in emotion recognition, particularly in human-computer interaction.

Recent Trends: You can identify emerging challenges and suggest future research directions in the context of your comprehensive study on the CREMA-D dataset [11].

III. Research Method

A. Data Set: CREMA-D dataset

The audio files in this dataset are named in such a way that the prefix letters describe the emotion classes as follows:

a = anger
d = disgust
f' = fear
h = happiness
n = neutral
sa = sadness
su = surprise

B. Data Visualisation and Exploration

The count of each emotion in our dataset is shown in Fig 1.

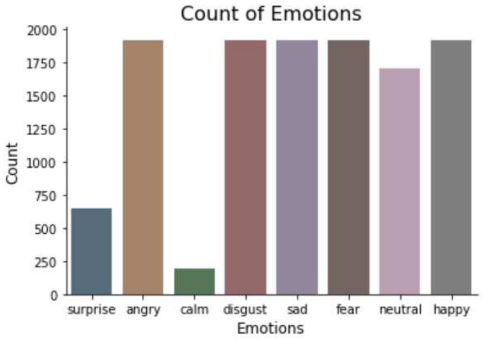

Fig 1: Data Visualisation and Exploration of data set

C. Wave plots:

Wave plots are graphical representations of audio signals that provide insight into the loudness of the audio at specific points in time. These plots display the amplitude of the audio signal along the time axis, allowing us to visualize how the loudness changes over the duration of the audio.[15] Waveforms are a valuable tool in audio analysis [12], offering a straightforward way to assess sound intensity and fluctuations. Waveplot for audio with fear and angry and happy emotions shown in Fig 2, 3 & 4.

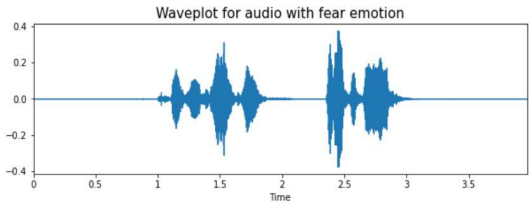

Fig 2: Wave plot for audio with fear emotions

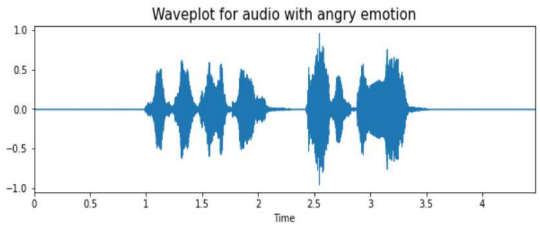

Fig 3: Wave plot for audio with angry emotions

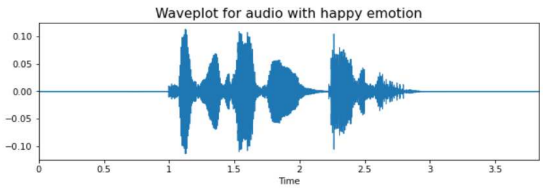

Fig 4: Wave plot for audio with happy emotions

A. Spectrograms:

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Spectrograms are visual representations of the frequency content of audio or other signals as they evolve over time. They provide a dynamic view of how different frequencies change with respect to time in audio or music signals. Spectrograms are created by dividing the audio signal into small time segments and then performing a Fourier transform on each segment to extract the frequency components [13]. These components are represented as a function of time, with intensity or colour indicating the magnitude of each frequency component. Both wave plots and spectrograms serve as essential tools in audio analysis, with wave plots conveying loudness information, and spectrograms providing a comprehensive view of frequency variations throughout an audio signal's duration. Spectrogram for audio with fear emotions shown in Fig 5,6 & 7.

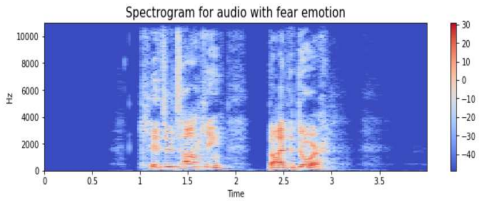

Fig 4: Spectogram for audio with fear emotions

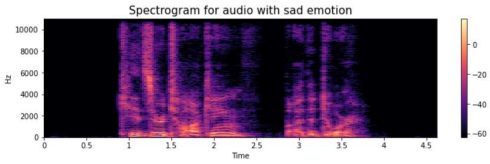

Fig 5: Spectogram for audio with fear emotions

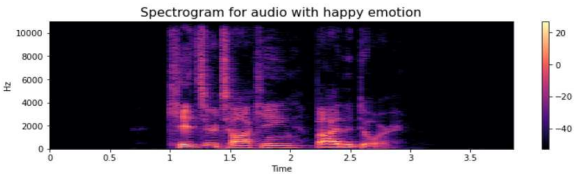

Fig 6: Spectogram for audio with fear emotions

B. Data Augmentation:

Data augmentation is a technique used to expand a dataset by generating new synthetic data samples through the introduction of small perturbations to the original training set. In the context of audio data, this involves applying various transformations such as noise injection, time shifting, pitch variation, and speed alteration. The primary objective of data augmentation is to enhance the model's ability to generalize by making it robust to these perturbations.

D. Simple Audio: Sample audio is loaded as shown in Fig 7.

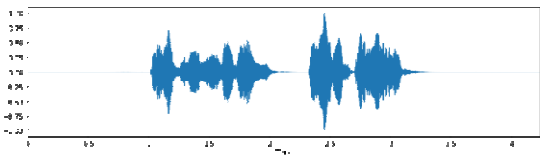

Fig 7: Sample audio

E. Noise Injection**:** Noise injection serves as a highly effective augmentation technique, providing assurance that our training model doesn't suffer from overfitting as shown in Fig 8.

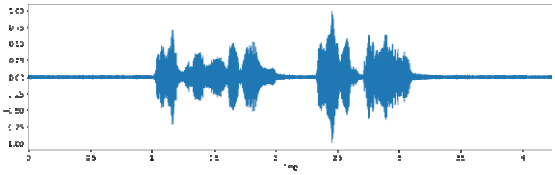

Fig 8: Noise Injection

F. Stretching: Stretching is shown in Fig 9 using stretch ().

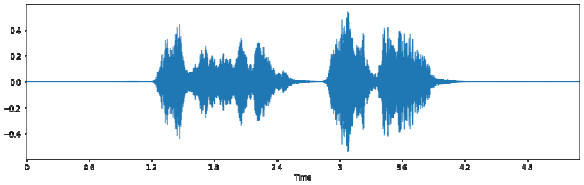

Fig 9: Stretching

G. Shifting**:** Shifting is shown in Fig 10 using shift ().

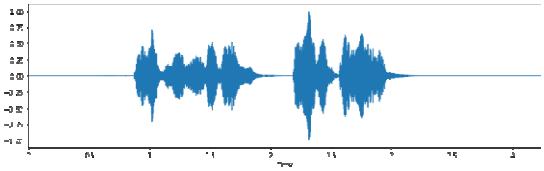

Fig 10: Shifting

H. Pitch**:** Pitching is shown in Fig 11 using pitch().

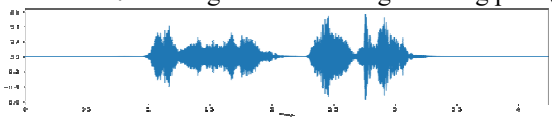

Fig 11: Pitching

I. Feature extraction is a crucial component in the analysis and establishment of relationships between various elements. Given that audio data, in its raw form, cannot be readily comprehended by models, it is essential to transform it into a format that is more interpretable. This is where feature extraction comes into play. Audio signals are inherently three-dimensional, with time, amplitude, and frequency acting as the three axes within this domain. Complete methodology is shown in Fig 12 and Fig 13.

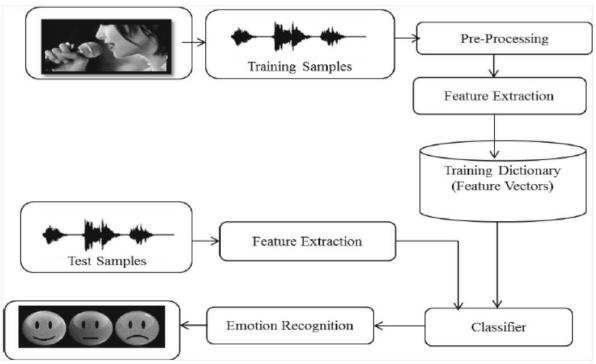

Fig 12: Methodology

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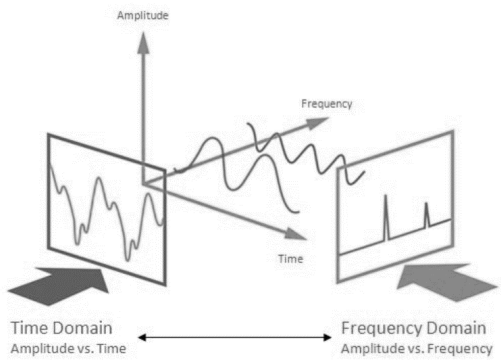

Fig 13: time, amplitude, and frequency

- Data Preparation

  Data is normalized and split data for training and testing multiclass classification is performed. Data splitting, Scaling and model compatibility values shown in Table 1:

Table 1: Data splitting, Scaling and model compatibility values

Data splitting	Scaling	Model compatibility
((27364, 162), (27364, 8), (9122, 162), (9122, 8))	((27364, 162), (27364, 8), (9122, 162), (9122, 8))	(27364, 162, 1), (27364, 8), (9122, 162, 1), (9122, 8))

IV. Results and Discussion

Accuracy of our model on test data is shown in Fig 14a and 14b.
9122/9122 [==========] - 1s 92us/step
Accuracy of our model on test data: 60.74%

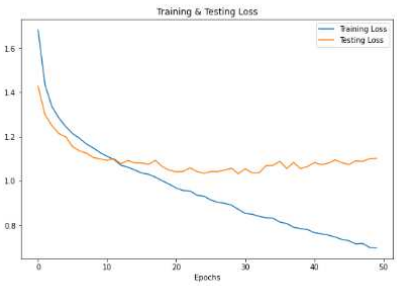

Fig 14 a) Training and Testing loss

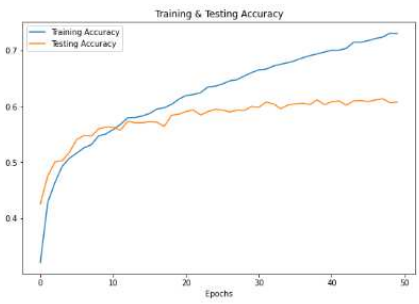

Fig 14 b) Training and Testing accuracy

Predicted and actual labels are shown in Table 2.

Predicted labels	Actual Labels
Neutral	Disgust
sad	sad
sad	sad
fear	disgust
happy	happy
sad	fear
disgust	sad
angry	happy
happy	happy

Based on these labels confusion matrix is shown below in fig 15.

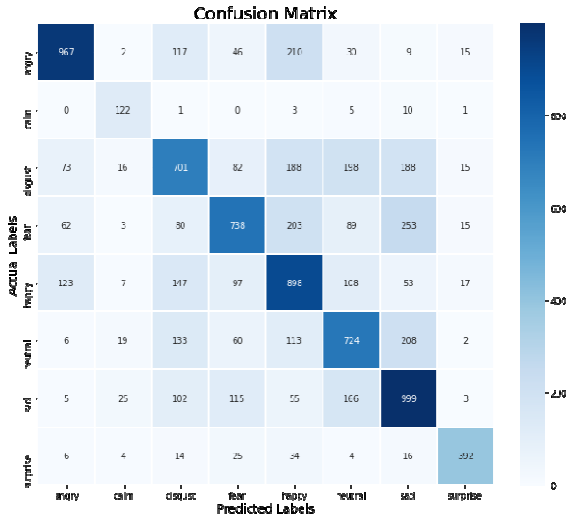

Fig 15: Predicted and actual labels

Summary of precision, recall, f score and support are sho wn in Table 2.

Table 3: Summary of precision, recall, f score & support

	precision	recall	F1-score	support
angry	0.78	0.69	0.73	1396
calm	0.62	0.86	0.72	142
disgust	0.54	0.48	0.51	1461
fear	0.63	0.51	0.57	1443
happy	0.53	0.62	0.57	1450
neutral	0.55	0.57	0.56	1265
sad	0.58	0.68	0.62	1470
surprise	0.85	0.79	0.82	495
accuracy			0.61	9122
macro avg	0.63	0.65	0.64	9122
Weighted avg	0.61	0.61	0.61	9122

IV Conclusion

It's noticeable that the model exhibits higher accuracy in predicting emotions like "surprise" and "anger," which is logical as audio files associated with these emotions

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exhibit noticeable variations in aspects like pitch and speed, distinguishing them from other audio samples.In the final analysis, we achieved an overall accuracy of 61% on our test data, which is a respectable result. However, further enhancements can be made by implementing additional augmentation techniques and exploring alternative feature extraction methods to improve our model's performance.

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